

# Polynomials – 4

Real roots: Descartes, Sturm and Newton's inequalities

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## Theoretical minimum.

**Theorem 1 (Descartes' rule of signs).** The number of positive roots of a real polynomial, counted with multiplicity, does not exceed the number of sign changes in the sequence of its non-zero coefficients, and has the same parity. Applying it to  $P(-x)$  bounds the number of negative roots.

**Theorem 2 (Sturm).** For a squarefree  $P$  let  $P_0 = P$ ,  $P_1 = P'$  and  $P_{k+1} = -\text{rem}(P_{k-1}, P_k)$ . The number of distinct real roots in  $(a, b]$  equals  $V(a) - V(b)$ , where  $V(x)$  is the number of sign changes in  $P_0(x), P_1(x), \dots$ . This is an exact count, not a bound.

**Fact 1 (Newton's inequalities).** If all roots of  $\sum_k \binom{n}{k} c_k x^k$  are real, then  $c_k^2 \geq c_{k-1} c_{k+1}$  for all  $k$ ; in particular the normalised coefficient sequence is log-concave and hence unimodal. The converse is false, so this is a necessary condition only.

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1. Prove Descartes' rule and Newton's inequalities, and deduce that a real-rooted polynomial has log-concave coefficients. Give a polynomial with log-concave coefficients which is not real-rooted.
  2. (Putnam 2014) Show that for every positive integer  $n$  all the roots of the polynomial  $\sum_{k=0}^n 2^{k(n-k)} x^k$  are real.
  3. (ФКН, весна 2023) Prove that for all natural  $n$  and  $p$  with  $0 < n \leq p$  all the roots of  $P_{n,p}(x) = \sum_{j=0}^n \binom{p}{j} \binom{p}{n-j} x^j$  are real.
  4. (VJIMC 2022) Assume that a real polynomial  $P(x)$  has no real roots. Prove that  $Q(x) = P(x) + \frac{P''(x)}{2!} + \frac{P^{(4)}(x)}{4!} + \dots$  has no real roots either.

5. (IMC) Let  $P$  be a polynomial of degree  $n$  with real coefficients and only real zeros.

- a) Prove that  $(n-1)(P'(x))^2 \geq nP(x)P''(x)$  for every real  $x$ .
- b) Examine the cases of equality.